19 In a box there are only red beads, green beads, yellow beads and pink beads.

The table shows each of the probabilities that, when a bead is taken at random from the box, the colour of the bead is red or is green.

Colour	red	green	yellow	pink
Probability	0.16	0.2		

The number of yellow beads is the same as the number of pink beads.

Vera is going to take at random one bead from the box and put the bead back in the box.

(a) Work out the probability that Vera will take a pink bead.

$1 - 0.16 - 0.20 = 0.64$

1 mark
Remaining two colours are equally likely.

$P(\text{pink}) = 0.64 \div 2 = 0.32$

1 mark
0.32.

(2)

Cathy is going to take a bead from the box.
She will record the colour of the bead and put the bead back in the box.
Cathy will do this 50 times.

(b) Work out an estimate for the number of times she will take a red bead from the box.

$\text{Expected red} = 0.16 \times 50$

1 mark
Probability × trials.

$= 8 \text{ times}$

1 mark
8.

(2)

(Total for Question 19 is 4 marks)

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